

Experiment 3-Medium Wind: Summary Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1990	11.141	13.692	21.601	15.0%	30.3%	36.2%	7.667	18.886	4.704	10.197	0.413	0.972	1.284
Proposed	1956	2.561	1.068	5.755	47.4%	77.2%	85.1%	15.374	29.454	7.938	17.139	0.557	1.299	1.496
PPO-GRU	1947	3.539	1.455	10.802	30.7%	61.3%	70.4%	14.956	27.364	7.863	17.915	0.557	1.210	1.459
MPC	1065	5.941	2.904	16.357	2.5%	17.5%	53.8%	2.693	3.674	1.483	2.668	0.140	0.142	0.178

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1990	99.5%	10	0.5%	0.5%	10	100.0%	0	0
Proposed	2000	1956	97.8%	44	2.2%	2.2%	44	100.0%	0	0
PPO-GRU	2000	1947	97.3%	53	2.6%	2.7%	53	100.0%	0	0
MPC	2000	1065	53.2%	935	46.8%	46.8%	935	100.0%	0	0

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	935	929	99.4%	617	66.4%	11.611
Proposed	935	906	96.9%	156	17.2%	2.775
PPO-GRU	935	903	96.6%	311	34.4%	3.937